# Substitution models, spectral decay and SatuTe

Rate matrices (A, C, G, T), spectral components, probability trajectories and SatuTe

## a JC

Equal exchangeabilities; equal frequencies

$$Q = \frac{1}{\mu}\begin{pmatrix} -3\alpha & \alpha & \alpha & \alpha \\ \alpha & -3\alpha & \alpha & \alpha \\ \alpha & \alpha & -3\alpha & \alpha \\ \alpha & \alpha & \alpha & -3\alpha \end{pmatrix}$$

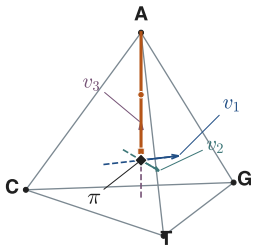
$$\mu = 3\alpha$$

Exact JC spectrum

$$\pi = (\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4})^{\mathsf{T}}$$

$$\lambda_1 = \lambda_2 = \lambda_3 = -\frac{4\alpha}{\mu} = -\frac{4}{3}$$

One decay rate; multiplicity 3




Straight trajectory to the stationary distribution

## b K80

Transitions α; transversions β

$$Q = \frac{1}{\mu}\begin{pmatrix} -s & \beta & \alpha & \beta \\ \beta & -s & \beta & \alpha \\ \alpha & \beta & -s & \beta \\ \beta & \alpha & \beta & -s \end{pmatrix}$$

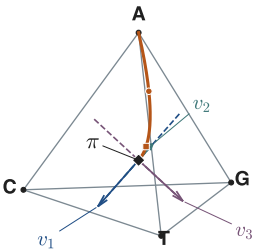
$$s = \alpha + 2\beta, \quad \mu = s$$

Illustrative parameter values

$$\pi = (0.25, 0.25, 0.25, 0.25)^{\mathsf{T}}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (-0.800, -1.600, -1.600)$$

Two decay rates; multiplicities 1 + 2




Curved trajectory with two decay rates

## c GTR

Unequal exchangeabilities and frequencies

$$Q = \frac{1}{\mu}\begin{pmatrix} -s_A & a\pi_C & b\pi_G & c\pi_T \\ a\pi_A & -s_C & d\pi_G & e\pi_T \\ b\pi_A & d\pi_C & -s_G & f\pi_T \\ c\pi_A & e\pi_C & f\pi_G & -s_T \end{pmatrix}$$

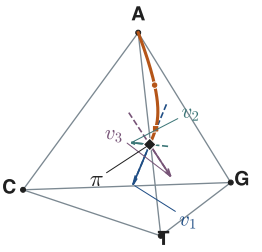
$$s_i = \textstyle\sum_{j \neq i} r_{ij}\pi_j, \quad \mu = \textstyle\sum_i \pi_i s_i$$

Illustrative parameter values

$$\pi = (0.35, 0.15, 0.25, 0.25)^{\mathsf{T}}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (-0.634, -1.334, -2.055)$$

Three distinct decay rates




Curved trajectory with three decay rates

$$Q^{\mathsf{T}} = V\ \mathrm{diag}(0, \lambda_1, \lambda_2, \lambda_3)V^{-1}, \qquad V = [\,\pi\ v_1\ v_2\ v_3\,]$$

$$p(t) = \pi + \textstyle\sum_{i=1}^{3} c_i e^{\lambda_i t} v_i, \qquad \lambda_0 = 0, \quad \lambda_1, \lambda_2, \lambda_3 < 0$$

p(t), initial state A ● t = 0.5 ■ t = 2 ◆ stationary distribution π

Dashed arrows show eigenvector directions at π; lengths are scaled for display.

SatuTe uses the same left eigenvectors and relative persistence

$$v_i = h_i, \qquad Q^{\mathsf{T}} h_i = \lambda_i h_i$$

$$w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}, \qquad w_1(t) = 1$$

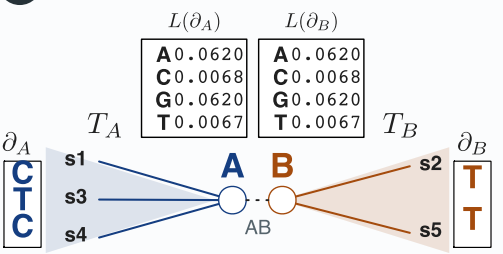
Maximum-likelihood tree inference

## d Alignment and site pattern

| | | ∂ | | |
|---|---|---|---|---|
| s1 | ... ATTAC | C | TTAC ATTAC | ... |
| s2 | ... CTTAT | T | TTAT CTTAC | ... |
| taxa s3 | ... AATAG | T | ATAG ATTAC | ... |
| s4 | ... ATTAG | C | TTAC ATTAC | ... |
| s5 | ... ATTAG | T | TCAC GTCAC | ... |

site pattern

## e Branch likelihood at AB



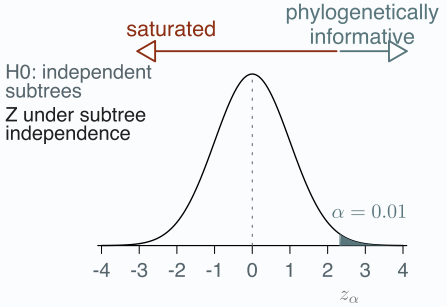

coherence coefficients

$$C_{\partial i} = \begin{pmatrix} L(\partial_A) \\ P(\partial_A) \end{pmatrix} \cdot h_i \begin{pmatrix} L(\partial_B) \\ P(\partial_B) \end{pmatrix} \cdot h_i$$

data re-evaluation

SatuTe

## g Decision under the null




reject independence when $Z_\lambda(t) > z_\alpha$

## f Weighted coherence and z-score

relative persistence

$$w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$$

weighted coherence

$$C^{\lambda}_{\partial}(t) = \textstyle\sum_{i=1}^{d} w_i(t) C_{\partial i}$$

alignment average

$$\widehat{C}_\lambda(t) = \textstyle\sum_{\partial} \frac{n_\partial}{n} C^{\lambda}_{\partial}(t)$$

$$Z_\lambda(t) = \frac{\sqrt{n}\,\widehat{C}_\lambda(t)}{\widehat{\sigma}_\lambda(t)}$$

$$H_0 : \ \widehat{C}_\lambda(t) \dot{\sim} \mathcal{N}(0, \hat{\sigma}^2_\lambda(t)/n)$$

Illustrated GTR

t = 1

| | |
|---|---|
| mode 1 | 1.00 |
| mode 2 | 0.50 |
| mode 3 | 0.24 |